Tutorial 13

Inner product spaces

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1. Let V be an inner product space and let $\mathbf{u}, \mathbf{v} \in V$. For each of the following statements, say whether it is true or false and give a reason for your answer.

   (a) If $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, they are orthogonal.

   > **Solution:** False. If $V = \mathbb{R}^2$, with the dot product as inner product, the vectors $\mathbf{u} = (1, 0)$ and $\mathbf{v} = (1, 1)$ are linearly independent, but not orthogonal.

   (b) If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, they are linearly independent.

   > **Solution:** The statement is false if one of the vectors is the zero vector. If both vectors are non-zero, the statement is true (see Theorem 5.2.5 in the Notes).

   (c) If $\mathbf{w}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, it is orthogonal to every vector in $\mathrm{span}(\{\mathbf{u}, \mathbf{v}\})$.

   > **Solution:** True. If $\mathbf{x} \in \mathrm{span}\{\mathbf{u}, \mathbf{v}\}$, then there are scalars λ, μ such that $\mathbf{x} = \lambda\mathbf{u} + \mu\mathbf{v}$. Then $\langle \mathbf{x}, \mathbf{w} \rangle = \langle \lambda\mathbf{u} + \mu\mathbf{v}, \mathbf{w} \rangle = \lambda\langle \mathbf{u}, \mathbf{w} \rangle + \mu\langle \mathbf{v}, \mathbf{w} \rangle = \lambda 0 + \mu 0 = 0$.

2. For any $p(x), q(x) \in P_n$, put

$$\langle p(x), q(x) \rangle = \int_{-1}^{1} p(x)q(x)\,dx.$$

   (a) Show that this defines an inner product on P_n.
   Do the check that $\langle p(x), p(x) \rangle = 0 \Rightarrow p(x) = \mathbf{0}$ for the case $n = 1$; for larger n it gets a bit messy, but the idea is the same.

   > **Solution:** The check is similar to that in Example 5.1.4 in the notes. For the last part, suppose $\langle p(x), p(x) \rangle = 0$, for some $p(x) = ax + b \in P_1$. Then
   >
   > $$0 = \langle p(x), p(x) \rangle = \int_{-1}^{1} (ax + b)^2\,dx = \frac{2}{3}a^2 + 2b^2.$$
   >
   > This implies that $a = b = 0$, and hence $p(x)$ is the zero polynomial.

   (b) Let $n = 3$ and $p(x) = x^3 - 1$. Find $\|p(x)\|$.

Solution:

$$\|p(x)\|^2 = \langle p(x), p(x)\rangle = \int_{-1}^{1} (x^3-1)^2\,dx = \frac{16}{7}, \text{and hence } \|p(x)\| = \frac{4}{\sqrt{7}}.$$

(c) Let $p(x) = x^3 - 1$ and $q(x) = x^3 + 1$. Determine $\|q(x)\|$, $\langle p(x), q(x)\rangle$, and the angle between $p(x)$ and $q(x)$. The *angle* between $p(x)$ and $q(x)$ is defined to be θ, where

$$\cos\theta = \frac{\langle p(x), q(x)\rangle}{\|p(x)\|\|q(x)\|}.$$

Solution:

$$\|q(x)\|^2 = \langle q(x), q(x)\rangle = \int_{-1}^{1} (x^3+1)^2\,dx = \frac{16}{7}, \text{and hence } \|q(x)\| = \frac{4}{\sqrt{7}}.$$

$$\langle p(x), q(x)\rangle = \int_{-1}^{1} (x^3-1)(x^3+1)\,dx = -\frac{12}{7}.$$

Hence

$$\cos\theta = \frac{-\frac{12}{7}}{\frac{4}{\sqrt{7}}\cdot\frac{4}{\sqrt{7}}} = -\frac{3}{4},$$

and therefore,

$$\theta \approx 2.418858$$

3. Let V be an inner product space and suppose that $\boldsymbol{u}, \boldsymbol{v}$ and $\boldsymbol{w}$ are vectors such that

$$\langle \boldsymbol{u}, \boldsymbol{v}\rangle = 2,\ \langle \boldsymbol{v}, \boldsymbol{w}\rangle = -3,\ \langle \boldsymbol{u}, \boldsymbol{w}\rangle = 5, \|\boldsymbol{u}\| = 1, \|\boldsymbol{v}\| = 2, \|\boldsymbol{w}\| = 7$$

Evaluate each of the following using this information:

(a) $\langle \boldsymbol{u}+\boldsymbol{v}, \boldsymbol{v}+\boldsymbol{w}\rangle$

Solution: 8 (Use Definition 5.1.1 and Theorem 5.1.5)

(b) $\langle \boldsymbol{u}-\boldsymbol{v}-2\boldsymbol{w}, 4\boldsymbol{u}+\boldsymbol{v}\rangle$

Solution: -40 (Use Definition 5.1.1 and Theorem 5.1.5)

(c) $\|2\boldsymbol{w} - \boldsymbol{v}\|$.

Solution: $\|2\boldsymbol{w} - \boldsymbol{v}\| = \sqrt{\langle 2\boldsymbol{w} - \boldsymbol{v}, 2\boldsymbol{w} - \boldsymbol{v}\rangle} = 2\sqrt{53}$.

4. Let

$$A = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ and } C = A \cup \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

(a) Show that A is an orthogonal set in $\mathbb{R}^4$, with its usual inner product. Is it an orthonormal set?

Solution: Check that the inner product of the two vectors in A is 0. The set is not orthonormal, since both vectors have norm equal to $\sqrt{2}$, and not 1.

(b) Use the Gram-Schmidt procedure to find an orthonormal basis B for span(C).

Solution: Since the two vectors in A are already orthogonal, it is enough to normalize them. Put

$$\boldsymbol{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \boldsymbol{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

Now let $\boldsymbol{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$ and

$$\begin{aligned} \boldsymbol{w}_3 &= \boldsymbol{v}_3 - \langle \boldsymbol{v}_3, \boldsymbol{u}_1 \rangle \boldsymbol{u}_1 - \langle \boldsymbol{v}_3, \boldsymbol{u}_2 \rangle \boldsymbol{u}_2 \\ &= \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{pmatrix} \end{aligned}$$

Finally normalise $\boldsymbol{w}_3$ to obtain $\boldsymbol{u}_3 = \sqrt{2}\boldsymbol{w}_3$. Then $B = \{\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{u}_3\}$.

5. Let A be a non-empty subset of an inner product space V. Define

$$A^{\perp} = \{\boldsymbol{v} \in V \ : \ \langle \boldsymbol{v}, \boldsymbol{u} \rangle = 0 \text{ for } \textbf{all } \boldsymbol{u} \in A\}.$$

(a) We start with an example. Let $V = \mathbb{R}^3$, with its usual inner product. Find $A^{\perp}$ if (i) $A = \{(0, 0, 1)\}$, (ii) $A = \{(1, 0, 0), (0, 0, 1)\}$.

Solution: (i) $A^{\perp}$ is the xy-plane. (ii) $A^{\perp}$ is the y-axis.

(b) In general (i.e. not only in the cases looked at in (a)), is $A^{\perp}$ always a subspace of V? Justify your answer.

Solution: Yes.We'll do the scalar multiplication check, and leave the other two to you. Suppose $\boldsymbol{w} \in A^{\perp}$, and k is a scalar. For any $\boldsymbol{u} \in A$, we have

$$\langle k\boldsymbol{w}, \boldsymbol{u} \rangle = k\langle \boldsymbol{w}, \boldsymbol{u} \rangle = k.0 = 0.$$

So $k\boldsymbol{w} \in A^{\perp}$, as required.

(c) Is it possible to have $A^{\perp} = V$? If so, when will this happen?

Solution: Yes, if $A = \{\mathbf{0}\}$.

(d) If U is a **subspace** of V, find $U \cap U^{\perp}$.

Solution: $U \cap U^{\perp} = \{\mathbf{0}\}$.

6. Let $C = \begin{pmatrix} 4 & -1 \\ -1 & 4 \end{pmatrix}$ and for $\boldsymbol{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \quad \boldsymbol{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{R}^2$ define $\langle \boldsymbol{u}, \boldsymbol{v} \rangle$ by the matrix product

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \begin{pmatrix} u_1 & u_2 \end{pmatrix} C \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

(a) Show that this defines an inner product on $\mathbb{R}^2$. [Hint: Do the checks of conditions 2 and 3 in the definition (those involving addition and scalar multiplication) *without* multiplying out the righthand side; this goes more quickly.]

Solution: Let $\boldsymbol{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, $\boldsymbol{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$, $\boldsymbol{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$ be elements of $\mathbb{R}^2$, and k a scalar (real number). Then

$$\begin{aligned} \langle \boldsymbol{u}, \boldsymbol{v} \rangle &= \begin{pmatrix} u_1 & u_2 \end{pmatrix} C \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \\ &= 4u_1v_1 - u_1v_2 - u_2v_1 + 4u_2v_2 \\ &= \begin{pmatrix} v_1 & v_2 \end{pmatrix} C \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \\ &= \langle \boldsymbol{v}, \boldsymbol{u} \rangle \end{aligned}$$

For the second property we use the distributive law for matrix multiplication:

$$\begin{aligned} \langle \boldsymbol{u} + \boldsymbol{v}, \boldsymbol{w} \rangle &= \langle \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \rangle \\ &= \begin{pmatrix} u_1 + v_1 & u_2 + v_2 \end{pmatrix} C \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \\ &= \begin{pmatrix} u_1 & v_1 \end{pmatrix} C \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} + \begin{pmatrix} v_1 & v_2 \end{pmatrix} C \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \\ &= \langle \boldsymbol{u}, \boldsymbol{w} \rangle + \langle \boldsymbol{v}, \boldsymbol{w} \rangle. \end{aligned}$$

The check that $\langle k\boldsymbol{u}, \boldsymbol{w} \rangle = k\langle \boldsymbol{u}, \boldsymbol{w} \rangle$ is similar and also uses properties of matrix multiplication.

For the last property note that

$$\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \rangle = 4u_1^2 - 2u_1u_2 + 4u_2^2 = 4(u_1 - \frac{u_2}{4})^2 + \frac{15}{4}u_2^2$$

(complete the square), so $\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \rangle \geq 0$ for all $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \in \mathbb{R}^2$. Also $\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \rangle = 0$ iff $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Use this inner product (not the usual one on $\mathbb{R}^2$) to answer the next two questions

(b) Find $\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|$.

Solution: $\left| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right| = \sqrt{\langle \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \end{pmatrix} \rangle} = \sqrt{4(4)^2 - 2(4)(1) + 4(1)^2} = \sqrt{60}.$

(c) Find all vectors in $\mathbb{R}^2$ that are orthogonal (perpendicular) to $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ (i.e. those whose inner product with this vector gives 0).

> **Solution:** $\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \perp \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ iff $\langle \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, \begin{pmatrix} 4 \\ 1 \end{pmatrix} \rangle = 0$. Working this out for this inner product gives $15u_1 = 0$. So all vectors of the form $\begin{pmatrix} 0 \\ c \end{pmatrix}$, $c \in \mathbb{R}$ are perpendicular to $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$.

(d) (A bit of a challenge) Can we replace the matrix C by any non-zero 2×2 matrix and still get an inner product? Give a reason for your answer.

> **Solution:** No. For the first property to be satisfied, the matrix C need to be symmetric. The last property need not even be satisfied for symmetric matrices, as the matrix $C = \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}$ shows.